IGSCA Benchmarking

March 26, 2026

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LLM Disclosure

The following describes the use of large language models (LLMs) in the preparation of this document:

- LLMs were used to assist with code drafting, debugging, and editing prose.
- All LLM-generated content was reviewed and verified by the author(s).
- Final responsibility for the accuracy and integrity of this work rests with the author(s).

Benchmarking

Computational Environment

Repository Info

```
git_hash <- system("git rev-parse HEAD", intern = TRUE)
git_short <- system("git rev-parse --short HEAD", intern = TRUE)
git_branch <- system("git rev-parse --abbrev-ref HEAD", intern = TRUE)
git_msg <- system("git log -1 --format=%s", intern = TRUE)
```

This document was compiled from commit **05b4345a** on branch **igsca-finetuning** (trying to fix documentation). Full SHA: 05b4345a1b45233689704fd615bb7bdd6e4b5570

```
sessioninfo::session_info()
```

```
─ Session info ───────────────────────────────────────────────────────────
 setting  value
 version  R version 4.5.1 (2025-06-13)
```

```
 os       Ubuntu 22.04.5 LTS
 system   x86_64, linux-gnu
 ui       X11
 language en_CA:en
 collate  en_US.UTF-8
 ctype    en_US.UTF-8
 tz       America/Toronto
 date     2026-03-26
 pandoc    3.6.3 @ /usr/share/positron/resources/app/quarto/bin/tools/x86_64/
(via rmarkdown)
 quarto   1.9.36 @ /opt/quarto/bin/quarto

─ Packages ───────────────────────────────────────────────────────────────────
 package     * version date (UTC) lib source
 bench       * 1.1.4   2025-01-16 [1] RSPM
 cli           3.6.5   2025-04-23 [1] RSPM
 digest        0.6.39  2025-11-19 [1] RSPM
 evaluate      1.0.5   2025-08-27 [1] RSPM
 fastmap       1.2.0   2024-05-15 [1] RSPM
 glue          1.8.0   2024-09-30 [1] RSPM
 grateful    * 0.3.0   2025-09-04 [1] RSPM (R 4.5.0)
 htmltools     0.5.8.1 2024-04-04 [1] RSPM
 htmlwidgets   1.6.4   2023-12-06 [1] RSPM
 jsonlite      2.0.0   2025-03-27 [1] CRAN (R 4.5.1)
 knitr         1.50    2025-03-16 [1] CRAN (R 4.5.1)
 lifecycle     1.0.4   2023-11-07 [1] RSPM (R 4.5.0)
 lobstr      * 1.1.2   2022-06-22 [1] RSPM
 pillar        1.11.1  2025-09-17 [1] RSPM
 profvis     * 0.4.0   2024-09-20 [1] RSPM
 rlang       * 1.1.6   2025-04-11 [1] RSPM
 rmarkdown     2.30    2025-09-28 [1] RSPM
 sessioninfo * 1.2.3   2025-02-05 [1] RSPM
 vctrs         0.6.5   2023-12-01 [1] RSPM (R 4.5.0)
 xfun          0.54    2025-10-30 [1] RSPM
 yaml          2.3.11  2025-11-28 [1] CRAN (R 4.5.1)

 [1] /home/mstruong/R/x86_64-pc-linux-gnu-library/4.5
 [2] /opt/R/4.5.1/lib/R/library
 * ── Packages attached to the search path.

──────────────────────────────────────────────────────────────────────────────
```

Packages Used

```
grateful::cite_packages(output = "paragraph", out.dir = ".")
```

We used R v. 4.5.1 [1] and the following R packages: bench v. 1.1.4 [2], cSEM v. 0.6.1.9000 [3], cSEM.DGP v. 0.0.0.9000 [4], devtools v. 2.4.6 [5], lavaan v. 0.6.21 [6], [7], lobstr v. 1.1.2 [8], profvis v. 0.4.0 [9], rmarkdown v. 2.30 [10], [11], [12], sessioninfo v. 1.2.3 [13], tidyverse v. 2.0.0 [14].

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